Fiscal Year:	FY 2020	**Task Last Updated:**	FY 12/16/2020
PI Name:	Sharma, Tasneem Ph.D.		
Project Title:	The Ex-Vivo Human Translaminar Autonomous System to Study SANS Pathogenesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	tpsharma@iu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	817-735-5108
Organization Name:	University of North Texas Health Science Center		
PI Address 1:	Department of Pharmacology & Neuroscience		
PI Address 2:	3500 Camp Bowie Blvd IREB 532		
PI Web Page:			
City:	Fort Worth	**State:**	TX
Zip Code:	76107	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Focused Investigations
Start Date:	06/01/2019	**End Date:**	08/31/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2020 (original end date was 5/31/2020) per TRISH (Ed., 8/21/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-FIP0013		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Focused Investigation Project During space travel, astronauts have to deal with long term exposure to zero gravity, ionizing cosmic rays, hypoxia, etc. The adverse effects of these challenging conditions result in a baffling physiological disorder such as Spaceflight Associated Neuro-ocular Syndrome (SANS). To understand the progression of SANS, in vitro human cells-based 3D biomimetic models capable of mimicking some of these adverse ocular symptoms is highly desired. This project aims to: 1) Differentiate retinal ganglion cells from human commercially-cleared induced pluripotent stem cells (iPSCs) based on published protocols and characterize phenotypes; 2) Adapt optimal differentiation protocols for spheroid neuro-glial co-culture based on experience with other human cell types and; 3) Establish a 3D co-culture system that mimics optic nerve tissue by directing axonal outgrowth within a micropatterned hydrogel.		

Rationale for HRP Directed Research:

There is an unmet clinical need for a new preclinical human model that can target disease etiology ex-vivo using intraocular pressure (IOP)/intracranial (ICP) to mimic the in-vivo paradigm of Spaceflight Associated Neuro-ocular Syndrome (SANS). We have successfully developed a two-chamber Translaminar Autonomous System (TAS) (U.S. patent application number 16/395610) that can independently regulate IOP and ICP using human donor posterior eyes. We can maintain tissue viability, integrity, complexity, and retinal functionality within the ex-vivo TAS model. The Translational Research Institute for Space Health (TRISH) funding allowed our research to have a high impact through a variety of modalities:
A) Impact on development of research model. Modifications performed on the TAS model to increase accuracy, efficiency, and productivity of research data.

1. We generated a new model design that allows a 16-chamber system and capacity to analyze 8 donor eyes within a single experimental run.

2. We built individualized pumps for each ICP and IOP chamber to modulate pressures with precision.

3. We utilized Arduino boards for automation of perfusion pumps to increase accuracy of pressure maintenance.

B) Impact on new prototype technologies: Development of the TAS Metrics software to utilize the ARKit platform to enable virtual measure using a mobile device's camera feature. High resolution photographic images can be captured of each posterior globe before and after each experimental run. The analyzed measures will include width and depth of the globe.

C) Scholarly Impact.

Patents: Utility patent application filed on April 26, 2019 under U.S. Patent Application No. 16/395,610. (already filed before TRISH funding but impacted further development) https://

Publications: In preparation for submission and data collected from TRISH funds

Research Impact/Earth Benefits:

1. The Ex-Vivo Human Translaminar Autonomous System to Study SANS Pathogenesis. Stacy M. Curry, Husain Lohawala, Gaurav Sharma, Tasneem P. Sharma.

2. Large scale application of the Ex-Vivo Human Translaminar Autonomous System. Husain Lohawala, Gaurav Sharma, Tasneem P. Sharma

D) Media and TRISH awareness impact.

1. https://

2. https://

E) Impact on prospective future collaborations.

Potential collaborations made at NASA's Human Research Program (HRP) annual Investigators' Workshop (IWS) and through various TRISH meetings:

1. Dr. Afshin Beheshti - miRNA project - NSPIRES program grant

2. Dr. Xiao wen Mao- radiation- Space Biology, HRP Impact funding mechanisms: Ability to apply for solicitations to NASA or other consortium based Human research project grants (BRASH 2101, NSPIRES awards) due to preliminary data generated from TRISH funds. We propose: Proposal 1: We will utilize the TAS model to study various simulated SANS pathogenic microenvironments using induced pluripotent stem cells: i. Aim 1) To determine effects of chronic mildly elevated ICP. ii. Aim 2) To determine effects of environmental factors. iii. Aim 3) To identify effective countermeasure strategies. Proposal 2: We will use conditioned medium from the TAS model and rodent biospecimens (undergoing basal, ground control, and flight-terminated research) to identify and compare pathogenic biomarkers from various simulated SANS pathogenic microenvironments: i. Aim 1) To characterize biomarkers within the IOP and ICP chamber conditioned medium of the TAS model. ii. Aim 2) To characterize biomarkers within spines and cerebral spinal fluid of rodents. iii. Aim 3) To contrast biomarkers from IOP and ICP chamber conditioned medium of the TAS model to spines and cerebral spinal fluid of rodents

Original project aims/objectives:
Spaceflight Associated Neuro-ocular Syndrome (SANS) is now considered an unexplained major risk factor for long-duration spaceflight. We employed our ex-vivo translaminar autonomous system (TAS) model (U.S. patent application number 16/395610) and human donor eyecups to recreate the ex-vivo SANS microenvironment. We proposed:

Specific Aim 1: To characterize the pathological changes occurring within the human posterior eyecup due to elevated IOP/ICP or optic nerve (ON) tilt.

Specific Aim 2: To determine the transport and functional capacity of retinal ganglion cells (RGCs) within the human posterior eyecup after elevated IOP/ICP or ON tilt.

Key findings:

1. We successfully maintained pressures for 14 days in all 6 different groups.

2. Visual micrograph depiction of posterior globes pre and post culture in the TAS model presented a general bulging at the optic nerve hypoplasia (ONH) for all groups.

3. We developed TAS metrics, a software, for automation of width and height measurements of globe pre and post culture in the TAS model. A greater difference or shrinkage of tissue was observed post culture for all groups compared to control. Diameter of the globe decreased but depth was larger for groups 1 and 4 post culture.

Task Progress:	4. We observed survival of RGCs by positive expression of retinal markers within all groups after TAS model culture. Increased expression of RBPMS (RNA-Binding Protein With Multiple Splicing) was observed in control samples. However, group 4 had elevated expression of all other retinal markers compared to control. The apoptotic and inflammatory markers were highly elevated for all groups compared to control. In addition, Groups 4 and 5 had the highest increase in glial fibrillary acidic protein (GFAP) expression. Specifically, group 4 had the most elevated BAX (apoptotic) and TLR4 (inflammatory) expression. 5. Increased extracellular matrix (ECM) protein secretion was depicted in ICP and IOP chamber conditioned medium of experimental groups. Compared to control, all groups displayed an increased trend in expression of fibronectin (FN) and collagen type IV (COLIV). Even though the same medium was perfused through both ICP and IOP chambers, but elevated levels of FN and COLIV were observed in the IOP chamber compared to ICP chamber. 6. Within control cross-sections, we observed normal and healthy axons compared to damaged or dying axons in the experimental groups. 7. Cupping and morphological reorganization of the ONH was detected in experimental groups. We observed increased thickening of laminar beams and restructuring of the ONH within experimental groups compared to control. Groups 4 and 5 also depicted slight tilting of the ON as expected. Further, extensive glial scarring and cupping was also observed in groups 4 and 5. An increase in inflammation was depicted by elevated levels of GFAP and TLR4 in all experimental groups. However, extensive TLR4 expression was detected in group 5. A basal level of TLR4 expression was observed within the control group. 8. Increased Terminal deoxynucleotidyl Transferase (TdT) dUTP Nick-End Labeling (TUNEL) positive cells were identified within ONH of experimental groups. High magnification micrographs of ONH depicted changes in morphology and remodeling. Maximal expression of TUNEL positive cells observed in group 1 compared to control and other groups. Remaining experimental groups did not show significant expression. 9. Minimal expression of cholera toxin subunit B (CTB) detected within ONH cross-sections between all groups, control and experimental. Slightly elevated expression was observed in group 3. 10. Increased retinal dysfunction was observed within specific experimental groups. Compared to normal, representative ERG responses (ERG) responses from other experimental groups showed varying amplitudes. Groups 2, 4, and 5 depicted waveforms that simulated rapid flicker. Impact of key findings on hypotheses, technology requirements, objectives, and specific aims of the original proposal: 1. Our TAS model will be the first ever ex-vivo human model to study SANS etiology via modulation of IOP, ICP, and ON tilt conditions. 2. We were able to successfully keep tissue viable in TAS and utilize the retina to analyze expression of retinal, apoptotic, and inflammatory markers. We observed a trend towards increased inflammation and apoptosis in groups 4 and 5. 3. The morphological and immunohistochemical alterations noted due to ICP and IOP changes were cupping and thickening of lamina at the ONH. 4. We could successfully access ON damage of RGCs under various group conditions. Our study will be the first to report survival and analysis of human donor eye tissue after prolonged periods in culture. 5. We successfully acquired electroretinograms from retinas post culture of human donor eyes, which pays credence to successful survival of retina within the model. 6. The development of TAS metrics software allowed automation of width and height measurements of the globe pre and post culture in the TAS model. This will allow accurate assessment of globe flattening. Proposed research plan for the coming year: Plan 1: We will utilize the TAS model in conjunction with induced pluripotent stem cells. We will then identify the genetic, environmental, and ICP related pathogenic conditions causing SANS by testing our aims under various SANS pathogenesis microenvironments to determine the effects of chronic mildly elevated ICP (aim 1), to determine the effects of environmental factors (aim 2), and to identify effective countermeasure strategies within the TAS model (aim 3). Plan 2: We will utilize the TAS model with human posterior eyecups and rodent biospecimens (undergoing basal, ground control, and flight-terminated research paradigms) to identify biomarkers that are secreted within the cerebral spinal fluid under various SANS conditions by the following aims: Aim 1: To characterize biomarkers within the IOP and ICP chamber conditioned medium of the TAS model, Aim 2: To characterize biomarkers within spines and cerebral spinal fluid of rodents, Aim 3: To contrast biomarkers from the IOP and ICP chamber conditioned medium of the TAS model to spines and cerebral spinal fluid of rodents.
Bibliography Type:	Description: (Last Updated: 08/17/2023)
Articles in Peer-reviewed Journals	Peng M, Curry SM, Liu Y, Lohawala H, Sharma G, Sharma TP. "The ex vivo human translaminar autonomous system to study spaceflight associated neuro-ocular syndrome pathogenesis." npj Microgravity. 2022 Oct 28;8:44. https://doi.org/10.1038/s41526-022-00232-5 ; PMID: 36307487; PMCID: PMC9616431 , Oct-2022